# ANTICHAIN SUBSET SUMS IN RANK-THREE MULTIPLICATIVE SEMIGROUPS

Abstract. Let $a, b, c \geq 2$ be pairwise coprime and $\mathcal{S} = \{a^k b^\ell c^m : k, \ell, m \geq 0\}$. For every real $1 < \rho < \min(a, b, c)$, every sufficiently large integer is a subset sum of $\mathcal{S} \cap [x, \rho x)$ for some $x$. Such a set is automatically a divisibility antichain, resolving Erdős Problem 123. For rational $\rho$ we prove the stronger statement that the Bernoulli subset sums of each large band satisfy a uniform local central limit theorem. The main new ingredient is an arithmetic rigidity argument on a triangular grid in exponent space; the remaining Fourier analysis is standard.

## 1. Statement and reduction

Fix pairwise-coprime integers $a, b, c \geq 2$, and write

$$\mathcal{S} = \{a^k b^\ell c^m : k, \ell, m \in \mathbb{Z}_{\geq 0}\}, \qquad d = \min(a, b, c).$$

Let $\rho = p/q \in \mathbb{Q}$ satisfy $1 < \rho < d$, with $p, q \in \mathbb{N}$ and $q > 0$. For an integer $x \geq 1$ set

$$\mathcal{B}_x = \mathcal{S} \cap [x, \rho x), \qquad W_x = \sum_{s \in \mathcal{B}_x} s, \qquad V_x^2 = \sum_{s \in \mathcal{B}_x} s^2,$$

and

$$\mu_x = \frac{W_x}{2}, \qquad \sigma_x = \frac{V_x}{2}.$$

Let $(\xi_s)_{s \in \mathcal{B}_x}$ be independent Bernoulli variables taking the values $0, 1$ with equal probability, and put $Y_x = \sum_{s \in \mathcal{B}_x} s\xi_s$.

**Theorem 1.1** (Local limit theorem)**.** *As $x \to \infty$ through the positive integers,*

$$\mathbb{P}(Y_x = n) = \frac{1}{\sqrt{2\pi}\,\sigma_x} \exp\left(-\frac{(n - \mu_x)^2}{2\sigma_x^2}\right) + o\left(\frac{1}{\sigma_x}\right), \tag{1}$$

*uniformly for $n \in \mathbb{Z}_{\geq 0}$. In particular, for all sufficiently large $x$, every integer $n$ with $|n - \mu_x| \leq \sigma_x$ is a subset sum of $\mathcal{B}_x$.*

**Theorem 1.2** (Erdős Problem 123)**.** *For every real $1 < \rho < d$, every sufficiently large integer $N$ can be written as*

$$N = \sum_{s \in A} s, \qquad A \subset \mathcal{S} \cap [x, \rho x)$$

*for some positive integer $x$. Consequently no two distinct elements of $A$ divide one another.*

The last assertion is immediate. If $u < v$ belong to $\mathcal{S}$ and $u \mid v$, unique factorization in the three pairwise-coprime bases gives $v/u = a^r b^s c^t \geq d$; two elements of $[x, \rho x)$ have ratio less than $\rho < d$.

We prove Theorem 1.1 for rational $\rho$, then obtain the theorem for an arbitrary real $\rho$ by choosing a rational $\rho'$ with $1 < \rho' < \rho$.

## 2. A triangular grid in the exponent slab

Put

$$\alpha = \log a, \qquad \beta = \log b, \qquad \gamma = \log c, \qquad \eta = \log \rho,$$

and, with $L = \log x$,

$$\mathcal{E}_L = \{(k, \ell, m) \in \mathbb{Z}_{\geq 0}^3 : L \leq \alpha k + \beta \ell + \gamma m < L + \eta\}.$$

This set parametrizes $\mathcal{B}_x$. The ratios of any two of $\alpha, \beta, \gamma$ are irrational: for instance $\alpha/\beta = r/s \in \mathbb{Q}$ would give $a^s = b^r$, contrary to $\gcd(a, b) = 1$.

We use the following elementary consequence of irrational rotation. If $u, v > 0$, $u/v \notin \mathbb{Q}$, and $0 < \eta < v$, then there is $R$ such that, whenever

$$uX + vY = A + \frac{\eta}{2}, \qquad X, Y \geq R,$$

there are nonnegative integers $r, s$, each within $R$ of the corresponding real coordinate, for which

$$A \le ur + vs < A + \eta. \tag{2}$$

Indeed, bounded multiples of $u/v$ form a finite $\eta/(8v)$-net modulo 1. Writing $r_0 = \lfloor X \rfloor$ and $y = (A - ur_0)/v$, choose a bounded integer $h$ so that the fractional part of $y - hu/v$ lies between $1 - 5\eta/(8v)$ and $1 - 3\eta/(8v)$; then take $r = r_0 + h$ and $s = \lceil y - hu/v \rceil$. This gives (2), and the displayed linear relation bounds the error in the second coordinate.

Applying this to the three pairs of logarithms gives a constant $R_0$ with the following property: every point $(X, Y, Z)$ on

$$\alpha X + \beta Y + \gamma Z = L + \frac{\eta}{2} \tag{3}$$

whose positive coordinates are at least $2R_0$, and of which at most one is zero, lies within $R_0$ coordinatewise of a point of $\mathcal{E}_L$; a zero coordinate remains zero. When all coordinates are positive, round $Y$ to a nearest integer $\ell_0$, replace $Z$ by $Z + (\beta/\gamma)(Y - \ell_0)$, and apply (2) to $\alpha, \gamma$. If one coordinate is zero, apply the two-variable statement directly to the other two.

**Proposition 2.1** (Band geometry)**.** *There are constants $c, C > 0$ and integers $D, K \ge 1$ such that, for all large $x$,*

$$cL^2 \le |\mathcal{B}_x| \le CL^2, \qquad cxL \le V_x \le CxL, \qquad cxL^2 \le W_x \le CxL^2. \tag{4}$$

*Moreover, for some $n \asymp L$ there are distinct elements $s_v \in \mathcal{B}_x$ indexed by*

$$\Delta_n^\circ = \{(i, j, r) \in \mathbb{Z}_{\ge 0}^3 : i + j + r = n\} \setminus \{(n, 0, 0), (0, n, 0), (0, 0, n)\},$$

*such that a zero coordinate of $v$ gives a zero corresponding exponent of $s_v$, and adjacent vertices satisfy*

$$A_{vw} s_v = B_{vw} s_w, \qquad 1 \le A_{vw}, B_{vw} \le K. \tag{5}$$

*Proof.* Choose $\tau_0$ so large that $\tau_0 / \max(\alpha, \beta, \gamma) > 4R_0$. For large $L$, let

$$n = \left\lfloor \frac{L + \eta/2}{\tau_0} \right\rfloor, \qquad \tau = \frac{L + \eta/2}{n};$$

then $n \asymp L$ and $\tau_0 \le \tau \le 2\tau_0$. For $v = (i, j, r) \in \Delta_n^\circ$, round the point

$$\left( \frac{\tau i}{\alpha}, \frac{\tau j}{\beta}, \frac{\tau r}{\gamma} \right)$$

to an exponent triple in $\mathcal{E}_L$. The corners were removed, so the rounding lemma applies. It preserves the coordinate faces. Distinct vertices have one target coordinate separated by more than $4R_0$, hence have distinct images. Adjacent targets differ by a bounded amount, so their rounded exponents differ coordinatewise by at most some fixed $D$. Clearing the positive and negative exponent differences gives (5) with $K = (abc)^D$.

Since $|\Delta_n^\circ| \asymp n^2$, this gives $|\mathcal{B}_x| \gg L^2$. Conversely, $k, \ell \ll L$, while for fixed $k, \ell$ there is at most one $m$, because two choices would change the logarithmic weight by at least $\gamma > \eta$. Thus $|\mathcal{B}_x| \ll L^2$. Finally $x \le s < \rho x$ throughout the band, and (4) follows by summing $s$ and $s^2$. □

## 3. Low-energy frequencies

For $t \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$, put

$$Q_x(t) = \sum_{s \in \mathcal{B}_x} \|st\|^2, \qquad \|y\| = \min_{m \in \mathbb{Z}} |y - m|.$$

The following is the arithmetic core of the proof.

**Proposition 3.1** (Rigidity)**.** *There are constants $z_0, C, \kappa, \delta > 0$ such that, for all large $x$ and $0 \le z \le z_0 L^2$,*

$$\operatorname{meas}\{t \in \mathbb{T} : Q_x(t) \le z\} \le \frac{1}{x} \exp\Big( C \left( 1 + \frac{z}{L} \right) \log(L + 2) \Big). \tag{6}$$

*The constants may be chosen so that $\delta d \le 1/8$ and*

$$Q_x(t) < \kappa L \quad \Longrightarrow \quad \|t\| \le \frac{\delta}{x}. \tag{7}$$

*Proof.* Use the grid from Proposition 2.1, and choose $0 < \delta_0 < \min(1/(4K), 1/32)$. For each vertex let $m_v$ be a nearest integer to $s_v t$, and call $v$ bad if $|s_v t - m_v| > \delta_0$. If there are $H$ bad vertices, then

$$H\delta_0^2 \le Q_x(t). \tag{8}$$

For $n/3 \le q \le 2n/3$, consider the row

$$R_q = \{(i, q-i, n-q) : 0 \le i \le q\}.$$

There are $\gg n$ disjoint such rows, so one contains at most $4H/n$ bad vertices. From its central half consider the disjoint paths

$$P_j = \{(q-j+r, j, n-q-r) : 0 \le r \le n-q\}, \qquad q/4 \le j \le 3q/4.$$

One of them contains at most $7H/n$ bad vertices. Their union is connected, has three endpoints on the three coordinate faces, and contains only $O(1 + H/n)$ bad vertices. If $8H < n$, the row and path can both be chosen with no bad vertices.

Orient the edges of this row and path. For an edge $vw$, define the integer

$$e_{vw} = B_{vw} m_w - A_{vw} m_v,$$

where (5) is used. If both endpoints are good, then

$$|e_{vw}| \le B_{vw}|m_w - s_w t| + A_{vw}|m_v - s_v t| < 2K\delta_0 < 1,$$

so $e_{vw} = 0$. Thus, when $Q_x(t) \le z$, only $O(1 + z/L)$ edge values can be nonzero, by (8) and $n \asymp L$. The row, path, positions of the nonzero edge values, and their bounded integer values can therefore take at most

$$\exp\Big(C\left(1 + \frac{z}{L}\right)\log(L+2)\Big) \tag{9}$$

possible values.

Fix these data and compare two compatible frequencies. If $\Delta_v$ denotes the difference of their nearest integers, then

$$B_{vw}\Delta_w = A_{vw}\Delta_v, \qquad B_{vw}s_w = A_{vw}s_v.$$

Connectivity gives $\Delta_v = rs_v$ for one rational number $r$. The three end weights respectively omit $a, b, c$; pairwise coprimality therefore gives their gcd, and hence the gcd of all weights on the row and path, as 1. Consequently $r \in \mathbb{Z}$. At a fixed root of weight $s_0$, the nearest integer lies between 0 and $s_0$, so a fixed choice of the edge data permits at most three root values modulo $s_0$. Each root value confines $t$ to an interval of length $1/s_0 \le 1/x$. Multiplying this by (9) proves (6).

If $Q_x(t) < \kappa L$ and $\kappa$ is sufficiently small, then (8) gives $8H < n$. Choose a completely good row and path. Every $e_{vw}$ is zero, so $m_v/s_v$ is constant. Its denominator divides the gcd of the three face weights, which is 1; hence the common ratio is an integer. At any vertex, $|s_v t - m_v| \le \delta_0$ and $s_v \ge x$, so initially $\|t\| \le \delta_0/x$. To sharpen the constant, apply this conclusion to a fixed rational subband of width at most 3/2. Writing $t = r + u$, we then have $|u| \le 1/(32x)$; nearest-integer rounding vanishes on that subband, and hence $Q_x(t) \ge u^2 \sum s^2 \gg x^2 L^2 u^2$. Thus $Q_x(t) < \kappa L$ forces $|u| \le \delta/x$ for any fixed $\delta > 0$ once $x$ is large. Choose $\delta d \le 1/8$ to obtain (7). □

## 4. Fourier analysis and completion of the proof

We shall use

$$|\cos(\pi y)| \le \exp(-2\|y\|^2), \tag{10}$$

which follows from $\sin(\pi u/2) \ge u$ for $0 \le u \le 1/2$.

**Lemma 4.1** (Fourier tails)**.**

$$\int_{T/V_x \le |t| \le 1/2} \left|\prod_{s \in \mathcal{B}_x} \cos(\pi s t)\right| dt = o\left(\frac{1}{V_x}\right), \qquad T = L^{1/4}. \tag{11}$$

*Proof.* For large $x$, $T/V_x < 2\delta/x$. On $T/V_x \le |t| \le 2\delta/x$, every $s \in \mathcal{B}_x$ satisfies $|st| < 2\delta d \le 1/4$, so (10) and $V_x^2 = \sum s^2$ give

$$\prod_{s \in \mathcal{B}_x} |\cos(\pi s t)| \le \exp(-2V_x^2 t^2).$$

Its integral is at most $V_x^{-1}\int_T^\infty e^{-2u^2}\,du = o(1/V_x)$.

On $2\delta/x \le |t| \le 1/2$, Proposition 3.1 gives $Q_x(t) \ge \kappa L$. By (10), the integrand is at most $e^{-2Q_x(t)}$. Layering by the values of $Q_x$ and using (6) yields

$$\int e^{-2Q_x(t)}dt \le \frac{2}{x}\int_{\kappa L}^{z_0L^2} e^{-2z+C(1+z/L)\log(L+2)}\,dz + e^{-2z_0L^2}.$$

For large $L$ this is $O(x^{-1}L^Ce^{-3\kappa L/2}) + e^{-2z_0L^2}$, which is $o(1/(xL)) = o(1/V_x)$ by (4). □

*Proof of Theorem 1.1.* Fourier inversion gives

$$\mathbb{P}(Y_x = n) = \int_{-1/2}^{1/2} e^{-2\pi i(n-\mu_x)t}\prod_{s\in\mathcal{B}_x}\cos(\pi st)\,dt. \tag{12}$$

On $|t| \le T/V_x$, all $|\pi st|$ tend uniformly to zero. The elementary estimates

$$|\cos u - e^{-u^2/2}| \le u^4, \qquad \left|\prod_j x_j - \prod_j y_j\right| \le \sum_j |x_j - y_j| \quad (|x_j|, |y_j| \le 1)$$

give

$$\left|\prod_{s\in\mathcal{B}_x}\cos(\pi st) - \exp\left(-\frac{\pi^2V_x^2t^2}{2}\right)\right| \le \pi^4t^4\sum_{s\in\mathcal{B}_x}s^4. \tag{13}$$

Since $s < px$ and $\sum s^2 = V_x^2$, $\sum s^4 \le (px)^2V_x^2$. Integrating the right side of (13) over $|t| \le T/V_x$ gives

$$O\left(\frac{(px)^2T^5}{V_x^3}\right) = o\left(\frac{1}{V_x}\right)$$

by (4). The Gaussian tail outside this interval is also $o(1/V_x)$. Hence the principal part of (12) equals

$$\int_{\mathbb{R}} e^{-2\pi i(n-\mu_x)t}\exp\left(-\frac{\pi^2V_x^2t^2}{2}\right)dt + o\left(\frac{1}{V_x}\right).$$

The standard Gaussian Fourier transform evaluates this integral as

$$\frac{\sqrt{2/\pi}}{V_x}\exp\left(-\frac{2(n-\mu_x)^2}{V_x^2}\right) = \frac{1}{\sqrt{2\pi}\,\sigma_x}\exp\left(-\frac{(n-\mu_x)^2}{2\sigma_x^2}\right).$$

The error is uniform in $n$, since the omitted oscillatory factor always has modulus 1. Lemma 4.1 proves (1).

If $|n-\mu_x| \le \sigma_x$, the main term in (1) is at least $e^{-1/2}/(\sqrt{2\pi}\sigma_x)$, whereas the error is $o(1/\sigma_x)$. Thus $\mathbb{P}(Y_x = n) > 0$ for all sufficiently large $x$, so $n$ is a subset sum of $\mathcal{B}_x$. □

*Proof of Theorem 1.2.* First suppose $\rho = p/q$ is rational. By Proposition 2.1, for all large $x$ the band has at least $4p^4$ elements, and hence

$$V_x^2 \ge 4p^4x^2, \qquad V_x \ge 2p^2x. \tag{14}$$

Also

$$W_{x+1} \le W_x + p^2(x+1). \tag{15}$$

Indeed, an element entering when $x$ is replaced by $x+1$ lies in an interval of length $p/q < p$, so there are at most $p$ such integers, each at most $p(x+1)$; elements leaving the band only decrease the sum.

Fix a sufficiently large initial $X$. For $N$ large enough that $W_X \le 2N$, choose the largest $x \ge X$ with $W_x \le 2N$. Such an $x$ exists and is finite, since every sufficiently large band is nonempty and then $W_x \ge x$. By maximality and (15),

$$0 \le 2N - W_x < W_{x+1} - W_x \le p^2(x+1) \le 2p^2x \le V_x.$$

Thus $|N - \mu_x| \le \sigma_x$, and Theorem 1.1 gives a subset of $\mathcal{B}_x$ summing to $N$.

For a real $1 < \rho < d$, choose a rational $\rho'$ with $1 < \rho' < \rho$. The rational case supplies the representation inside $[x, \rho'x) \subset [x, \rho x)$. As observed at the start, every such subset is a divisibility antichain. □